

Relativistic–Nonrelativistic Matching and Matter–Antimatter Constraints for Exotic Spin-Dependent Interactions: A Unified Framework and the Limits of One-Sided Bounds as CPT Tests

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(Dated: September 30, 2026)

Light, weakly coupled bosons can mediate exotic spin-dependent interactions over a wide range of distances. Experimental constraints on such interactions are commonly reported using the non-relativistic potential basis originating with Dobrescu and Mocioiu (DM), whereas Lorentz- and CPT-violating effects are naturally formulated within the Standard-Model Extension (SME). We establish a controlled correspondence between the spin-dependent coefficients of the minimal SME fermion sector and nonrelativistic operator structures by applying the field normalisation and Foldy–Wouthuysen (FW) procedure appropriate to derivative kinetic terms. The analysis covers b_μ , $d_{\mu\nu}$, $g_{\lambda\mu\nu}$ and $H_{\mu\nu}$, which are shown to be the complete set of minimal-SME fermion coefficients that generate spin-dependent nonrelativistic terms through third order in $1/m$; the dictionary is structural and does not identify SME coefficients with two-body DM couplings. The canonical nonrelativistic Hamiltonian reproduces the established spin and spin–momentum structures, including the mass-enhanced d_{i0} and $g_{[kl]0}$ contributions and the momentum-linear combination $d_{ij} + d_{00}\delta_{ij}$.

We then investigate whether published matter and antimatter upper limits can be converted into a quantitative CPT asymmetry. A signed quantity such as $A = (g_f - g_{\bar{f}})/(g_f + g_{\bar{f}})$ is meaningful only when g_f and $g_{\bar{f}}$ are signed observables or well-defined estimators. Experimental exclusion results are instead usually one-sided bounds on positive quantities such as $|g|$. If $U_f = s_f g_\star$ and $U_{\bar{f}} = s_{\bar{f}} g_\star$ describe two such limits on the same underlying coupling magnitude, their apparent asymmetry is exactly

$$A_U = \frac{U_f - U_{\bar{f}}}{U_f + U_{\bar{f}}} = \frac{s_f - s_{\bar{f}}}{s_f + s_{\bar{f}}},$$

which contains no CPT parameter. We therefore interpret this quantity as a sensitivity-gap diagnostic rather than as a CPT observable.

A compiled registry contains 283 constraint records, of which 247 are admitted to the present analysis. Twenty-two analysed records involve antimatter, and fifteen matter–antimatter pairs (thirteen independent) satisfy the adopted matching requirements in coupling convention, operator label, fermion sector, and interaction range. The resulting bound-based asymmetries span approximately 0.33–1 in magnitude, but their size is not evidence for CPT violation. The present coverage is strongly matter-dominated, with substantial deficits in electron–nucleus, neutron–nucleus, and proton–nucleus sectors. We formulate a practical two-stage programme in which balanced sensitivity regions are first identified through a sensitivity-ratio function and genuine CPT tests are subsequently performed using signed observables or full likelihood functions.

I. INTRODUCTION

Searches for weakly coupled new interactions at low energies provide a complementary route to high-energy searches for physics beyond the Standard Model (BSM). Light bosons, including axionlike particles and new spin-1 states, can mediate long-range forces whose effects may be probed through precision spectroscopy, atomic and molecular systems, torsion balances, comagnetometers, spin sensors, and antimatter experiments [1–4].

A standard phenomenological language for such interactions is provided by the sixteen rotationally invariant nonrelativistic structures introduced by Dobrescu and Mocioiu [2]. Modern treatments have clarified the relation between different historical potential conventions, reduced coupling constants, and experimental applications

[3]. The DM classification is, however, a nonrelativistic phenomenological framework rather than a complete relativistic effective field theory. In particular, the label of a DM potential does not by itself determine the Lorentz or CPT properties of an underlying interaction.

The SME provides a systematic effective-field-theory description of Lorentz and CPT violation [5, 6]. In its minimal fermion sector, coefficients such as b_μ , $H_{\mu\nu}$, and $d_{\mu\nu}$ modify the Dirac Lagrangian and produce observable nonrelativistic effects. The Foldy–Wouthuysen transformation provides the standard route from the relativistic Dirac description to a two-component nonrelativistic Hamiltonian [7, 8].

The relationship between these descriptions is useful but must be formulated carefully. An SME coefficient is a one-fermion background quantity, whereas a DM coupling such as $g_A^e g_A^N$ is a product of couplings at two interaction vertices. Consequently, an SME coefficient cannot in general be identified numerically with a DM coupling. A legitimate connection must proceed through the nonrel-

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ativistic operator structure and, where required, through an explicit two-fermion scattering amplitude.

This distinction is especially important for matter–antimatter comparisons. Antiprotonic helium spectroscopy has already constrained exotic spin-dependent interactions involving antiprotons [9], while positronium provides a theoretically clean leptonic matter–antimatter system for precision tests [10]. At the same time, spin-based quantum sensors and NV-diamond techniques continue to improve constraints on exotic interactions in ordinary matter [11–13]. These developments make a systematic comparison of matter and antimatter constraints timely.

A natural dimensionless quantity for signed couplings is

$$A_\alpha = \frac{g_\alpha^f - g_\alpha^{\bar{f}}}{g_\alpha^f + g_\alpha^{\bar{f}}}, \quad (1)$$

where f and $\bar{f}$ denote the corresponding matter and antimatter sectors. Equation (1) is meaningful when the two quantities are signed measurements or estimators with well-defined probability distributions. Most searches for weak exotic interactions instead publish one-sided exclusion limits on positive quantities such as $|g_\alpha|$. Applying Eq. (1) directly to such limits therefore changes its interpretation.

The central statistical result of this work is that the resulting quantity measures relative experimental sensitivity rather than CPT violation. This follows algebraically and does not depend on the details of the experimental data set.

The work has three objectives. First, we establish a controlled relativistic–nonrelativistic correspondence for the spin-dependent SME fermion coefficients. Second, we distinguish genuine signed matter–antimatter tests from comparisons of one-sided exclusion limits. Third, we use a compiled constraint registry to quantify present matter–antimatter coverage and identify experimental sectors in which matched measurements would be particularly informative.

The one-body dictionary constructed here is complete in a precise sense: for a single fermion species in the minimal SME, every spin-dependent term of the nonrelativistic Hamiltonian through third order in $1/m$ arises from b_μ , $d_{\mu\nu}$, $g_{\lambda\mu\nu}$ or $H_{\mu\nu}$, and all four are reduced here. We do not claim a two-body coupling dictionary: a true two-body correspondence requires explicit matching of relativistic amplitudes for specified mediator models. Similarly, the numerical asymmetries obtained from one-sided upper limits are not interpreted as evidence for new physics.

II. RELATIVISTIC AND NONRELATIVISTIC DESCRIPTIONS

A. The SME fermion sector

Unless otherwise stated, the field-theory equations in this section use natural units $\hbar = c = 1$ and metric signature $(+, -, -, -)$. To first order in the Lorentz-violating coefficients relevant here, the fermion Lagrangian may be written as

$$\mathcal{L} = \bar{\psi} \left(i\gamma^\mu \partial_\mu - m - b_\mu \gamma_5 \gamma^\mu - \frac{1}{2} H_{\mu\nu} \sigma^{\mu\nu} + \frac{i}{2} d_{\mu\nu} \gamma_5 \gamma^\mu \overleftrightarrow{\partial}^\nu + \frac{i}{4} g_{\lambda\mu\nu} \sigma^{\lambda\mu} \overleftrightarrow{\partial}^\nu \right) \psi, \quad (2)$$

where

$$\bar{\psi} \overleftrightarrow{\partial}^\nu \psi = \bar{\psi} \partial^\nu \psi - (\partial^\nu \bar{\psi}) \psi \quad (3)$$

and

$$\sigma^{\mu\nu} = \frac{i}{2} [\gamma^\mu, \gamma^\nu]. \quad (4)$$

The coefficient $g_{\lambda\mu\nu}$ is antisymmetric in its first two indices. The coefficients b_μ and $g_{\lambda\mu\nu}$ are CPT odd, whereas $H_{\mu\nu}$ and $d_{\mu\nu}$ are CPT even in the usual SME classification [5, 6, 14]. This classification should not, however, be used by itself to infer the sign of a particle–antiparticle energy shift. The relation between the particle and antiparticle Hamiltonians must be obtained from the complete relativistic theory.

The coefficients $d_{\mu\nu}$ and $g_{\lambda\mu\nu}$ are particularly delicate because they occur in the kinetic sector. Their time-derivative terms must first be removed from the canonical time evolution by an appropriate field redefinition. Performing a FW reduction before this canonical normalisation can lead to spurious sign differences or to a non-Hermitian intermediate Hamiltonian [8].

B. The Dobrescu–Mocioiu framework

For two nonrelativistic fermions separated by $\mathbf{r}$, the original DM construction identifies sixteen rotationally invariant structures formed from spin, relative momentum or velocity, $\hat{\mathbf{r}}$, and the interaction range. For a boson of mass m_ϕ the range is

$$\lambda = \frac{\hbar}{m_\phi c}. \quad (5)$$

The detailed normalisation of the potentials has appeared in several forms in the literature. In particular, the modern review of Cong *et al.* gives a complete field-theoretic formulation in terms of reduced coupling constants and explicitly discusses the differences between historical conventions [3]. We therefore use labels such as V_2 , V_{2+3} , V_{4+5} , V_{9+10} , and V_{12+13} only as database and

literature labels. We do not reproduce a second set of explicit potential formulae here, since doing so would introduce unnecessary convention dependence into the SME matching.

This point is important for the present work. The database may contain an experimental limit reported for a combined potential such as V_{2+3} , but that does not justify assigning the result to V_2 or V_3 separately. Combined experimental quantities are therefore retained in their published form.

C. One-body SME operators versus two-body potentials

The correct conceptual relation is

$$\begin{array}{c}
 \text{relativistic SME operator} \\
 \downarrow \\
 \text{one-body nonrelativistic operator} \\
 \downarrow \\
 \text{two-body interaction structure,}
 \end{array} \quad (6)$$

where the final step requires an explicit model for the source or mediator.

For example, b_i produces a one-body spin coupling. It is not itself the product of two mediator couplings. A two-body interaction involving two spins may have a related spin structure, but the numerical coefficient depends on the underlying relativistic interaction and on the source. The present work therefore establishes structural correspondence, not equality of SME coefficients and DM coupling constants.

III. CANONICAL FOLDY–WOUTHUYSEN REDUCTION

A. General procedure

The standard FW transformation separates the positive- and negative-energy sectors of the Dirac Hamiltonian and produces a two-component Hamiltonian as an expansion in $|\mathbf{p}|/m$ [7]. For Lorentz-violating kinetic terms, the canonical field redefinition must precede the FW transformation [8].

Schematically, after canonical normalisation one writes

$$H = \beta m + \mathcal{E} + \mathcal{O}, \quad [\beta, \mathcal{E}] = 0, \quad \{\beta, \mathcal{O}\} = 0, \quad (7)$$

and the ordinary FW expansion begins with

$$H_{\text{FW}} = \beta m + \mathcal{E} + \frac{\beta \mathcal{O}^2}{2m} - \frac{1}{8m^2} [\mathcal{O}, [\mathcal{O}, \mathcal{E}]] + \dots \quad (8)$$

For the free Dirac Hamiltonian,

$$\mathcal{O}_0 = \boldsymbol{\alpha} \cdot \mathbf{p}, \quad \mathcal{O}_0^2 = \mathbf{p}^2, \quad (9)$$

so that the positive-energy sector begins as

$$H_0 = m + \frac{\mathbf{p}^2}{2m} + \mathcal{O}\left(\frac{p^4}{m^3}\right). \quad (10)$$

The derivative coefficient $d_{\mu\nu}$ cannot be treated by simply replacing $i\partial_\mu$ with a classical momentum in the original Lagrangian. The time-derivative part changes the canonical normalisation of the field. The controlled treatment is therefore the one developed by Kostelecký and Lane [8].

B. b_μ

The b_μ interaction is

$$\mathcal{L}_b = -b_\mu \bar{\psi} \gamma_5 \gamma^\mu \psi. \quad (11)$$

Retaining the leading nonrelativistic terms from the canonical FW Hamiltonian gives

$$H_{\text{NR}}^{(b)} = -b_i \sigma^i + \frac{b_0}{m} \boldsymbol{\sigma} \cdot \mathbf{p} + \mathcal{O}\left(\frac{p^2}{m^2}\right). \quad (12)$$

Thus the spatial component gives a leading spin coupling of order m^0 , whereas b_0 contributes to a momentum-linear spin structure suppressed by one power of m .

The signs in Eq. (12) are those of the canonical nonrelativistic Hamiltonian of Ref. [8]. Here and in Eqs. (15), (17), (22) and Table I, the momentum components are the covariant ones used in that reference, $p_j = -p^j$, so that $\boldsymbol{\sigma} \cdot \mathbf{p} \equiv \sigma^j p_j$ and the cross products $\mathbf{p} \times \mathbf{H}_E$ and $\mathbf{p} \times \mathbf{g}_{00}$ are built from p_j . This reading is fixed by the free-particle term of the same derivation, $m\mathcal{P}_0 = -p_j \gamma^0 \gamma^j$, which reproduces $+\boldsymbol{\alpha} \cdot \mathbf{p} + \beta m$ only if $p_j = -p^j$. Written with the physical momentum p^j , every momentum-linear term below changes sign; the momentum-independent terms do not.

C. $H_{\mu\nu}$

The tensor interaction is

$$\mathcal{L}_H = -\frac{1}{2} H_{\mu\nu} \bar{\psi} \sigma^{\mu\nu} \psi. \quad (13)$$

Define

$$B_H^i = \frac{1}{2} \epsilon^{ijk} H_{jk}, \quad (H_E)^i = H_{0i}. \quad (14)$$

The leading nonrelativistic Hamiltonian is

$$H_{\text{NR}}^{(H)} = \mathbf{B}_H \cdot \boldsymbol{\sigma} - \frac{1}{m} \boldsymbol{\sigma} \cdot (\mathbf{p} \times \mathbf{H}_E) + \mathcal{O}\left(\frac{p^2}{m^2}\right). \quad (15)$$

The first term is a leading spin coupling, while the second is linear in momentum. Equation (15) follows directly from the canonical SME Hamiltonian and should not be replaced by a sign inferred only from the CPT label of $H_{\mu\nu}$.

D. $d_{\mu\nu}$

The derivative interaction is

$$\mathcal{L}_d = \frac{i}{2} d_{\mu\nu} \bar{\psi} \gamma_5 \gamma^\mu \overleftrightarrow{\partial}^\nu \psi. \quad (16)$$

After the required canonical field redefinition and FW transformation, the terms relevant through first order in momentum are

$$H_{\text{NR}}^{(d)} = m d_{i0} \sigma^i - (d_{ij} + d_{00} \delta_{ij}) p_j \sigma^i + \mathcal{O}\left(\frac{p^2}{m}\right). \quad (17)$$

The mass-enhanced term

$$H_{\text{NR}}^{(d_{i0})} = m d_{i0} \sigma^i \quad (18)$$

is the established result quoted in the SME nonrelativistic Hamiltonian [8].

The momentum-linear contribution is important because the physically relevant combination is not d_{ij} or d_{00} separately at this order, but

$$d_{ij} + d_{00} \delta_{ij}. \quad (19)$$

Accordingly, the earlier treatment in which the d_{ij} and d_{00} terms were assigned independent unresolved signs is replaced here by the canonically normalised result of Ref. [8]. The apparent sign differences arose from reading p_j as the physical momentum: the mass-enhanced d_{i0} term, which carries no momentum, agreed throughout, while only the momentum-linear terms appeared reversed.

E. $g_{\lambda\mu\nu}$

The remaining kinetic-sector interaction is

$$\mathcal{L}_g = \frac{i}{4} g_{\lambda\mu\nu} \bar{\psi} \sigma^{\lambda\mu} \overleftrightarrow{\partial}^\nu \psi, \quad (20)$$

which enters Γ^ν as $\frac{1}{2} g^{\lambda\mu\nu} \sigma_{\lambda\mu}$ [8]. Because its $\nu = 0$ components modify the time-derivative term, the same canonical field redefinition used for $d_{\mu\nu}$ is required before the FW reduction. The index structure separates $g_{\lambda\mu\nu}$ into four families, labelled by whether the antisymmetric pair is spatial-spatial ($[kl]$) or temporal-spatial ($[0k]$) and whether the derivative index is temporal (0) or spatial (j). Defining

$$\tilde{g}_j = \frac{1}{2} \epsilon_{jkl} g_{kl0}, \quad (\mathbf{g}_{00})_k = g_{k00}, \quad (21)$$

the terms relevant through first order in momentum are

$$H_{\text{NR}}^{(g)} = -m \tilde{\mathbf{g}} \cdot \boldsymbol{\sigma} - \boldsymbol{\sigma} \cdot (\mathbf{p} \times \mathbf{g}_{00}) - \frac{1}{2} \epsilon_{klm} g_{mlj} p_j \sigma^k + \mathcal{O}\left(\frac{p^2}{m}\right), \quad (22)$$

from the $g_{[kl]0}$, $g_{[0k]0}$ and $g_{[kl]j}$ families respectively. The $g_{[0k]j}$ family first contributes at $\mathcal{O}(p^2/m)$. The first term

is mass enhanced, like the d_{i0} term, because the temporal derivative index brings in the energy. Equation (22) reproduces the corresponding terms of Ref. [8] under the covariant momentum convention stated in Sec. III B; since that convention was fixed from the free-particle term alone, the agreement of the g sector is an independent check on it.

The coefficients e_μ , f_μ and $g_{\lambda\mu\nu}$ are dimensionless and do not arise from any renormalisable, gauge-invariant term of the full SME; they can be generated only through higher-dimensional operators after electroweak symmetry breaking and are therefore expected to be suppressed [6]. For protons and neutrons, however, strong binding may generate appreciable effective values [15], which makes the g sector relevant to the nucleon channels of the registry.

F. Completeness of the spin-dependent sector

In the nonrelativistic Hamiltonian of Ref. [8], carried to third order in $1/m$, every term that contains the spin operator involves only b_μ , $d_{\mu\nu}$, $g_{\lambda\mu\nu}$ or $H_{\mu\nu}$. The coefficients a_μ , $c_{\mu\nu}$ and e_μ appear only in spin-independent terms, and f_μ does not appear at linear order; it can in fact be removed entirely by a field redefinition [16]. Since every DM structure other than V_1 is spin dependent, the four coefficients treated above form the complete spin-dependent one-body dictionary for a single fermion species in the minimal SME at this order. Coefficients of higher mass dimension in the nonminimal SME lie outside this statement.

G. Particle-antiparticle transformation

A further point is essential for the matter-antimatter analysis. For the one-particle SME Hamiltonian, the corresponding antiparticle Hamiltonian can be written in the same form after the coefficient substitutions

$$\begin{aligned} a_\mu &\rightarrow -a_\mu, & b_\mu &\rightarrow +b_\mu, & c_{\mu\nu} &\rightarrow +c_{\mu\nu}, \\ d_{\mu\nu} &\rightarrow -d_{\mu\nu}, & e_\mu &\rightarrow -e_\mu, & f_\mu &\rightarrow -f_\mu, \\ g_{\lambda\mu\nu} &\rightarrow +g_{\lambda\mu\nu}, & H_{\mu\nu} &\rightarrow -H_{\mu\nu}. \end{aligned} \quad (23)$$

These substitutions are part of the canonical particle-antiparticle construction of the SME Hamiltonian [8].

Equation (23) also illustrates why the statement ‘‘CPT even’’ or ‘‘CPT odd’’ should not be used as a shortcut for the sign of an experimentally measured energy shift. The discrete-symmetry classification concerns the relativistic operator, whereas the particle-antiparticle relation of a particular nonrelativistic observable follows from the complete Hamiltonian and the conventions used for the antiparticle state.

H. Controlled matching summary

The controlled results are summarised in Table I. The last column deliberately describes operator structure rather than assigning a DM potential number to an SME coefficient.

The table is complete for the spin-dependent one-body sector: the remaining minimal-SME coefficients a_μ , $c_{\mu\nu}$, e_μ and f_μ produce no spin-dependent term at these orders (Sec. III F). It is not a two-body coupling dictionary: a true two-body matching requires an explicit relativistic scattering amplitude and cannot be inferred from the one-body Hamiltonian alone.

IV. MATTER–ANTIMATTER COMPARISON OF CONSTRAINTS

A. Signed observables

Let θ_f denote a signed observable associated with a fermion and let $\theta_{\bar{f}}$ be the corresponding observable for the antifermion. Once the complete relativistic transformation has been derived, exact CPT symmetry imposes a relation of the form

$$\theta_{\bar{f}} = \eta_{\text{CPT}} \theta_f, \quad \eta_{\text{CPT}} = \pm 1, \quad (24)$$

where the value of η_{CPT} depends on the particular signed observable and its conventions. It must therefore be derived rather than assigned solely from the name of the underlying SME coefficient.

A corresponding CPT residual is

$$\Delta_{\text{CPT}} = \theta_{\bar{f}} - \eta_{\text{CPT}} \theta_f. \quad (25)$$

If the measurements are approximately Gaussian, a simple test statistic is

$$\chi_{\text{CPT}}^2 = \frac{(\theta_{\bar{f}} - \eta_{\text{CPT}} \theta_f)^2}{\sigma_{\bar{f}}^2 + \sigma_f^2 - 2\eta_{\text{CPT}} \text{Cov}(\theta_f, \theta_{\bar{f}})}. \quad (26)$$

This is the appropriate type of construction when signed information and its uncertainty are available.

B. One-sided exclusion limits

The situation is different when the published results are

$$|g_f| < U_f, \quad |g_{\bar{f}}| < U_{\bar{f}}, \quad U_f, U_{\bar{f}} > 0. \quad (27)$$

One can define the purely numerical quantity

$$A_U = \frac{U_f - U_{\bar{f}}}{U_f + U_{\bar{f}}}, \quad (28)$$

but it should not be called a measured CPT asymmetry.

C. Sensitivity-gap theorem

Suppose two experiments constrain the same underlying coupling magnitude $g_\star$ with sensitivity factors s_f and $s_{\bar{f}}$:

$$U_f = s_f g_\star, \quad U_{\bar{f}} = s_{\bar{f}} g_\star. \quad (29)$$

Substitution into Eq. (28) gives

$$A_U = \frac{s_f - s_{\bar{f}}}{s_f + s_{\bar{f}}}. \quad (30)$$

The underlying coupling has cancelled. Equation (30) therefore contains no CPT parameter.

For example,

$$s_{\bar{f}} \ll s_f \implies A_U \rightarrow +1, \quad (31)$$

whereas

$$s_f \ll s_{\bar{f}} \implies A_U \rightarrow -1. \quad (32)$$

The result holds irrespective of whether the underlying signed interaction is CPT even or CPT odd. Consequently, a value $|A_U| \simeq 1$ obtained from two positive exclusion limits cannot by itself be interpreted as evidence for CPT violation.

There is also a useful algebraic warning concerning signed CPT-odd quantities. If exact CPT symmetry requires

$$g_{\bar{f}} = -g_f, \quad (33)$$

then the denominator of Eq. (1) vanishes. Exact sign reversal therefore does not correspond to $A = \pm 1$. The apparent saturation of A_U at ± 1 is a property of comparing positive magnitudes.

D. Sensitivity-ratio diagnostic

A more transparent quantity derived from two one-sided limits is

$$R_\alpha(\lambda) = \frac{U_{\bar{f},\alpha}(\lambda)}{U_{f,\alpha}(\lambda)}, \quad (34)$$

provided the limits have first been harmonised to the same coupling normalisation, confidence convention, and operator definition.

For a symmetric measure of the mismatch one may define

$$G_\alpha(\lambda) = |\ln R_\alpha(\lambda)|. \quad (35)$$

Equal sensitivity corresponds to

$$R_\alpha = 1, \quad G_\alpha = 0. \quad (36)$$

TABLE I. Controlled nonrelativistic structures obtained from the spin-dependent minimal-SME coefficients. The rows exhaust the spin-dependent terms of the minimal SME through the orders shown (Sec. III F). The table does not identify an SME coefficient with a two-body mediator coupling. Momentum components are covariant, $p_j = -p^j$, as in Ref. [8].

SME coefficient	CPT character	Nonrelativistic contribution	Leading order	Operator structure
b_i	odd	$-b_i \sigma^i$	m^0	spin-linear
b_0	odd	$(b_0/m) \boldsymbol{\sigma} \cdot \mathbf{p}$	m^{-1}	spin-momentum
H_{ij}	even	$\mathbf{B}_H \cdot \boldsymbol{\sigma}$	m^0	spin-linear
H_{0i}	even	$-m^{-1} \boldsymbol{\sigma} \cdot (\mathbf{p} \times \mathbf{H}_E)$	m^{-1}	spin-momentum
d_{i0}	even	$md_{i0} \sigma^i$	m^1	spin-linear
d_{ij}, d_{00}	even	$-(d_{ij} + d_{00} \delta_{ij}) p_j \sigma^i$	m^0	spin-momentum
$g_{[kl]0}$	odd	$-m \tilde{\mathbf{g}} \cdot \boldsymbol{\sigma}$	m^1	spin-linear
$g_{[0k]0}$	odd	$-\boldsymbol{\sigma} \cdot (\mathbf{p} \times \mathbf{g}_{00})$	m^0	spin-momentum
$g_{[kl]j}$	odd	$-\frac{1}{2} \epsilon_{klm} g_{mlj} p_j \sigma^k$	m^0	spin-momentum

Unlike A_U , G_α makes no claim to measure a signed physical coupling. It is explicitly a diagnostic of experimental coverage.

For future experiments that provide signed likelihoods $\mathcal{L}_f(g)$ and $\mathcal{L}_{\bar{f}}(g)$, the CPT-conserving likelihood can be written schematically as

$$\mathcal{L}_{\text{CPT}}(g) = \mathcal{L}_f(g) \mathcal{L}_{\bar{f}}(\eta_{\text{CPT}} g), \quad (37)$$

with the appropriate nuisance parameters and correlations included. This is fundamentally different from comparing the amplitudes of two exclusion curves.

V. EXPERIMENTAL CONSTRAINT DATABASE

A. Compilation strategy

To quantify the present experimental landscape, we constructed a structured registry of published constraints on exotic spin-dependent interactions. Each record contains, where available, the fermion sectors, coupling family, DM potential label, interaction range, and published constraint. The common interaction-range coordinate is

$$\lambda = \frac{\hbar}{m_\phi c}. \quad (38)$$

The registry contains

$$N_{\text{compiled}} = 283, \quad N_{\text{analysed}} = 247. \quad (39)$$

Thirty-six records are held out of the physics analysis. Twenty-eight are combined envelope curves, twelve in g_{AGV} and sixteen in g_{PGS} , labelled in the source database as combined or review-only and lacking sufficient information for a unique potential assignment. Five are the 90% confidence-level versions of the hydrogen bounds of Ref. [17], which duplicate the corresponding 95% CL curves. Two are contributor-template placeholders. One is an astrophysical electron-nucleon bound for which the available source information does not establish a unique potential assignment.

TABLE II. Constraint records in the compiled registry. The counts are registry entries, not statistically independent measurements.

Coupling family	Compiled	Analysed
g_{AGA}	69	66
g_{PGS}	53	37
g_{SGS}	49	49
g_{VGV}	45	44
g_{AGV}	37	25
g_{PGP}	17	15
V_1 unlabelled coupling	11	11
Template placeholder	2	0
Total	283	247

The analysed records are distributed among the principal coupling families as shown in Table II.

The registry includes constraints from atomic and molecular spectroscopy, torsion balances, comagnetometry, spin sensors, positronium, antiprotonic helium, and astrophysical observations. These platforms probe substantially different interaction ranges and spin sectors [3, 4, 9, 10, 12, 13].

B. Data-quality controls

The numerical registry requires several qualifications.

First, records may be filed under more than one coupling family. Consequently, the 247 analysed records do not represent 247 statistically independent measurements. Forty-eight records correspond to repeated entries of the same bound under different coupling labels, leaving 199 unique constraint curves for the purposes of the curve inventory. The term “unique” is used deliberately: these curves should not be treated as statistically independent unless their experimental origins are independently established.

Second, six registry entries contain interaction-range values between about 10^{23} and 10^{52} m. These values are physically implausible and are consistent with a unit-token parsing error. They are excluded from interpreta-

tions of global range coverage. Three of the matched pairs of Sec. VII A use an affected record: the g_{AGA} , V_3 , $e - e^+$ pair (matter side, Ref. [18]) and the two g_{AGA} , V_3 , $e - \bar{p}$ pairs that use Ref. [9] on the antimatter side. In each case the implausible tail lies outside the overlapping range on which the comparison is evaluated, $[2.5 \times 10^{-12}, 1.6 \times 10^{-5}]$ m, $[2.0 \times 10^{-11}, 1.0 \times 10^{-8}]$ m and $[2.0 \times 10^{-11}, 1.3 \times 10^{-8}]$ m respectively.

Third, several experiments report combined potential classes rather than individual components. Labels such as V_{2+3} and V_{4+5} are therefore retained as published. They are not decomposed into separate potential limits without an independent theoretical justification.

These controls are important because a compilation of exclusion curves can otherwise create an artificial impression of statistical precision.

VI. MATTER–ANTIMATTER COVERAGE

Of the 247 analysed records, 22 involve an antimatter sector: nine electron–antiproton records, six electron–antimuon records, five electron–positron records, one antiproton–helium record, and one molecular-ion record associated with an antimuon sector. All six electron–antimuon records come from muonium, which contains an antimuon; they are counted as antimatter for the same reason as the $dd\mu^+$ record. Thus

$$N_{\text{antimatter}} = 22, \quad N_{\text{matter}} = 225. \quad (40)$$

The relevant issue for a CPT comparison is not simply whether an antimatter measurement exists. A directly comparable result requires agreement in operator definition, fermion sector, coupling normalisation, and overlapping interaction range.

The sector inventory in the present registry is shown in Table III. Each record carries exactly one sector label; the counts are nevertheless not statistically independent, because the same bound can appear under several coupling families.

The proton–antinucleus entry requires particular care. Antiprotonic helium spectroscopy provides an antiproton–nuclear constraint [19], but the published result uses a normalisation that cannot be inserted into the corresponding proton–nucleus entries without an explicit conversion. It is therefore not counted as a directly comparable $p - \bar{N}$ result in the adopted coupling normalisation.

The three nuclear-source categories contain

$$50 + 49 + 18 = 117 \quad (41)$$

matter-sector registry records. The corresponding directly comparable antimatter count is zero in the adopted convention. The principal coverage limitation is therefore not the absence of matter-sector constraints but the lack of matched antimatter measurements in sectors where matter data are already numerous.

VII. DIRECT MATTER–ANTIMATTER COMPARISONS

A. Matched pairs

Applying the registry matching criteria to coupling family, potential label, fermion sector, and overlapping interaction range gives fifteen direct matter–antimatter pairs. Two of them repeat two others exactly: the source compilation files the same V_3 curves of Ref. [20] under both $g_V g_V$ and $g_P g_P$. The fifteen pairs therefore contain thirteen independent comparisons.

For each pair, the published curves were interpolated onto a common logarithmic range coordinate. A nominal 300-point grid was used in the original numerical implementation. The pointwise quantity was

$$A_U(\lambda_i) = \frac{U_f(\lambda_i) - U_{\bar{f}}(\lambda_i)}{U_f(\lambda_i) + U_{\bar{f}}(\lambda_i)}. \quad (42)$$

The values in Table IV are retained as diagnostic quantities only. They are not interpreted as estimates of a CPT parameter.

The fifteen values range from approximately 0.33 to 1. This broad range is precisely what is expected if the statistic is responding to different experimental sensitivities rather than to a common signed physical asymmetry.

B. Internal null behaviour

The $g_S g_S$, V_1 , electron–positron comparison is particularly instructive because it produces a large bound-based asymmetry, $|A_U| \simeq 0.87$, despite the fact that the quantity being compared is not a signed CPT-odd observable. The same procedure therefore produces large apparent asymmetries in channels where interpreting the result as a CPT signal is unjustified.

This is an internal null check on the statistic. The numerical value of A_U is controlled by the relative strengths of the two exclusion curves, not by the discrete-symmetry classification of the underlying interaction.

The lowest value in the table,

$$|A_U| \simeq 0.334, \quad (43)$$

also demonstrates that the statistic is not mathematically forced to be close to unity. Its magnitude simply tracks the degree of mismatch between the available constraints.

The four pairs that use the hydrogen bounds of Ref. [17] test this directly. Each has a counterpart with the same coupling, potential and antimatter partner but an older electron–proton bound on the matter side. In all four cases the newer, tighter matter-side bound gives the larger $|A_U|$: $0.9998 \rightarrow 0.9999$ (g_{AGA} , V_2 , with Ref. [9]), $0.8044 \rightarrow 1.0000$ (g_{AGA} , V_3 , with Ref. [20]), $0.8236 \rightarrow 0.9303$ (g_{AGA} , V_3 , with Ref. [9]) and $0.7994 \rightarrow 0.9304$ ($g_P g_P$, V_3 , with Ref. [9]). Nothing changed on the antimatter side, and the statistic moved with the sensitivity ratio, as Eq. (30) requires.

TABLE III. Matter and antimatter coverage by the canonical fermion-sector labels used in the registry. Counts are analysed registry records and are not statistically independent measurements. Nineteen records (sixteen astrophysical bounds and the $\mu - N$, $\bar{p}$ -He and $dd\mu^+$ entries) belong to no sector pair and are not shown.

Matter sector	Antimatter sector	Matter	Antimatter	Coverage
$e - e$	$e - e^+$	40	5	both
$e - p$	$e - \bar{p}$	17	9	both
$e - \mu$	$e - \bar{\mu}$	0	6	antimatter only (muonium)
$e - n$	$e - \bar{n}$	16	0	no antimatter record
$\mu - \mu$	$\mu - \bar{\mu}$	1	0	no antimatter record
$n - p$	$n - \bar{p}$	9	0	no directly matched record
$n - n$	$n - \bar{n}$	8	0	no antimatter record
$p - p$	$p - \bar{p}$	0	0	no record
$e - N$	$e - \bar{N}$	50	0	no directly matched record
$n - N$	$n - \bar{N}$	49	0	no directly matched record
$p - N$	$p - \bar{N}$	18	0	no directly matched record

TABLE IV. Bound-based matter-antimatter sensitivity diagnostics for the fifteen directly matched pairs in the registry, with the matter and antimatter sources. The quantity $|A_U|$ is the mean over the overlapping range, calculated from positive upper limits, and is therefore not a CPT-violation estimator. Bootstrap intervals are not quoted because the underlying uncertainty model does not provide formal confidence intervals for a CPT parameter.

Coupling	Potential	Sector	Matter / antimatter source	$ A_U $
$g_A g_A$	V_3	$e - \bar{p}$	[17] / [20]	1.0000
$g_A g_A$	V_2	$e - \bar{p}$	[17] / [9]	0.9999
$g_A g_A$	V_2	$e - \bar{p}$	[21] / [9]	0.9998
$g_A g_A$	V_2	$e - e^+$	[18] / [21]	0.9892
$g_A g_A$	V_3	$e - e^+$	[18] / [20]	0.9539
$g_P g_P$	V_3	$e - e^+$	[20] / [20]	0.9535
$g_V g_V$	V_3	$e - e^+$	[20] / [20]	0.9535
$g_P g_P$	V_3	$e - \bar{p}$	[17] / [9]	0.9304
$g_A g_A$	V_3	$e - \bar{p}$	[17] / [9]	0.9303
$g_S g_S$	V_1	$e - e^+$	[22] / [10]	0.8727
$g_A g_A$	V_3	$e - \bar{p}$	[20] / [9]	0.8236
$g_A g_A$	V_3	$e - \bar{p}$	[20] / [20]	0.8044
$g_P g_P$	V_3	$e - \bar{p}$	[20] / [9]	0.7994
$g_V g_V$	V_3	$e - \bar{p}$	[20] / [9]	0.7994
$g_A g_A$	V_2	$e - e^+$	[23] / [21]	0.3336

VIII. COVERAGE BY POTENTIAL LABEL

The same registry can be organised by its DM potential labels. The result is shown in Table V. Each record carries exactly one potential label. A bare numeric prefix in the source filenames denotes the potential itself, so a record labelled 3 bounds V_3 alone; only records reported for the combination are labelled V_{2+3} .

Seven of the twelve classified potential groups have no antimatter record despite nonzero matter coverage:

$$V_{11}, \quad V_{15}, \quad V_{16}, \quad V_{1a}, \quad V_{9+10}, \quad V_{12+13}, \quad V_{2+3}. \quad (44)$$

The V_3 group has the largest number of antimatter entries in the present registry. This is a statement about the existing experimental literature and not about an intrinsic suitability of this potential for a CPT test.

Recent experimental developments continue to improve sensitivity to spin-dependent interactions using quantum sensors and related techniques [3, 4, 11–13].

TABLE V. Matter and antimatter registry coverage by DM potential label (analysed records).

Potential label	Matter	Antimatter
V_{4+5}	58	3
V_{9+10}	31	0
V_3	29	10
V_{1a}	14	0
V_2	17	3
V_1	42	5
V_{12+13}	16	0
V_{11}	7	0
V_8	5	1
V_{15}	2	0
V_{16}	2	0
V_{2+3}	2	0

A recent torsion-balance proposal explicitly targets V_{4+5} and V_{12+13} interactions between polarized electrons and

unpolarised nucleons [24]. These developments show that matter-sector coverage continues to improve, while the present registry still shows a substantial antimatter deficit.

IX. STATISTICAL METHODOLOGY AND LIMITATIONS

A. Interpolation and correlation

The 300-point interpolation grid used for the matched curves should not be interpreted as 300 independent observations. Published exclusion curves are smooth functions of interaction range and neighbouring interpolated points are strongly correlated.

The exploratory analysis estimated the autocorrelation function of the interpolated residual sequence,

$$C(k) = \frac{\sum_i (x_i - \bar{x})(x_{i+k} - \bar{x})}{\sum_i (x_i - \bar{x})^2}, \quad (45)$$

and used the first crossing of $C(k) = e^{-1}$ as a characteristic correlation scale.

This procedure is useful as a diagnostic against treating every interpolated point as independent. It is not a rigorous derivation of the likelihood of the original experimental measurements, because the published curves do not generally provide the full covariance matrix.

Accordingly, the autocorrelation-based effective degrees of freedom are not used as the primary basis for a CPT significance claim.

B. Uncertainty assignment

Most published constraint curves do not provide point-by-point covariance matrices. The original analysis therefore estimated local fractional uncertainties σ_f , $\sigma_{\bar{f}}$ (between 2% and 50%) from the curvature of the curves in logarithmic coordinates. The weighted diagnostic compares the two curves directly,

$$\chi_w^2 = \sum_i \frac{[U_f(\lambda_i) - U_{\bar{f}}(\lambda_i)]^2}{\sigma_f^2 U_f^2(\lambda_i) + \sigma_{\bar{f}}^2 U_{\bar{f}}^2(\lambda_i)}, \quad (46)$$

and tests only whether the two exclusion curves coincide, i.e. the hypothesis of Eq. (48). For the autocorrelation diagnostic it is rescaled by n_{eff}/n before comparison with a χ^2 distribution on n_{eff} degrees of freedom.

A parametric bootstrap with 2000 resamples was then used to propagate the estimated uncertainty model to summary quantities. This is not a bootstrap of the original experimental events. It propagates an assumed uncertainty model inferred from published curves and therefore cannot restore systematic uncertainties or correlations that were not published.

Furthermore, taking an absolute value produces a non-linear statistic. An interval for $|A_U|$ therefore cannot be

interpreted as a confidence interval for a signed CPT parameter.

C. The correct null hypothesis

A conventional signed CPT test asks whether

$$H_0 : \theta_{\bar{f}} = \eta_{\text{CPT}} \theta_f. \quad (47)$$

The one-sided-bound comparison instead asks whether

$$H_0 : \frac{U_{\bar{f}}}{U_f} = 1. \quad (48)$$

These hypotheses are not equivalent. Consequently, even a very precise determination that $U_{\bar{f}}/U_f \neq 1$ is not a measurement of CPT violation.

X. A PRACTICAL FRAMEWORK FOR FUTURE CPT TESTS

A genuine matter-antimatter test of an exotic spin-dependent interaction requires four conditions.

First, the same physical operator must be defined on both sides of the comparison. Spin, momentum, velocity, and antiparticle-state conventions must be specified explicitly.

Second, the transformation between the matter and antimatter observables must be derived from the relativistic operator or scattering amplitude. The sign of an effective nonrelativistic Hamiltonian term cannot safely be inferred from a coupling label alone.

Third, the experiments must provide signed information or likelihood functions whenever a signed CPT test is intended. Two positive upper limits do not determine whether the underlying coupling is positive, negative, or zero.

Fourth, the matter and antimatter experiments must have sufficiently overlapping sensitivity in interaction range and operator convention. A comparison made where one experiment is many orders of magnitude less sensitive than the other is principally a comparison of experimental reach.

A. Stage I: sensitivity coverage

For each operator and range, calculate

$$R_\alpha(\lambda) = \frac{U_{\bar{f},\alpha}(\lambda)}{U_{f,\alpha}(\lambda)}. \quad (49)$$

A balanced interval may be defined by

$$|\ln R_\alpha(\lambda)| < \epsilon, \quad (50)$$

where ϵ is selected according to the intended experimental criterion.

This step identifies where matter and antimatter measurements have comparable sensitivity. It does not itself test CPT.

B. Stage II: likelihood-level comparison

Suppose a matter experiment provides a signed estimator $\hat{g}_f$ and likelihood $\mathcal{L}_f(g_f)$, while the corresponding antimatter experiment provides $\mathcal{L}_{\bar{f}}(g_{\bar{f}})$. The CPT-conserving hypothesis is

$$g_{\bar{f}} = \eta_{\text{CPT}} g_f. \quad (51)$$

A likelihood-ratio statistic can then be defined as

$$q_{\text{CPT}} = -2 \ln \frac{\max_g \mathcal{L}_f(g) \mathcal{L}_{\bar{f}}(\eta_{\text{CPT}} g)}{\max_{g_f, g_{\bar{f}}} \mathcal{L}_f(g_f) \mathcal{L}_{\bar{f}}(g_{\bar{f}})}. \quad (52)$$

In an actual analysis, nuisance parameters, correlated systematics, non-Gaussian likelihoods, and the precise statistical construction of each experiment must be retained. Equation (52) is therefore a framework rather than a numerical result of the present database.

XI. EXPERIMENTAL IMPLICATIONS

A. Electron–nucleus, neutron–nucleus, and proton–nucleus sectors

The registry contains substantial matter-sector coverage in

$$N_{e-N} = 50, \quad N_{n-N} = 49, \quad N_{p-N} = 18, \quad (53)$$

while the corresponding directly comparable antimatter counts are zero in the adopted coupling normalisation.

These sectors consequently provide clear targets for future matched matter–antimatter measurements. The objective should not simply be a stronger upper limit. The experiment should reproduce the same operator definition, coupling normalisation, and interaction-range dependence as an existing matter-sector measurement so that the two results can eventually be compared at the likelihood level.

B. Velocity-dependent and related potential classes

The registry contains matter-sector entries but no antimatter entries for V_{1a} , V_{9+10} , and V_{12+13} . Such gaps identify areas where a future antimatter measurement could add qualitatively new coverage.

The recent torsion-balance proposal of Jin *et al.* specifically targets V_{4+5} and V_{12+13} using polarized electrons and unpolarised nucleon sources [24]. The work illustrates how operator-specific sensitivity can be developed

experimentally, but a corresponding antimatter source or bound system would still be required for a direct CPT comparison.

C. Electron–positron and electron–antiproton systems

Electron–positron and electron–antiproton systems currently provide the most developed antimatter coverage in the registry. Positronium is particularly useful because its leptonic composition allows precision comparison with bound-state QED calculations [10]. Atomic spectroscopy has also been used to constrain exotic spin-dependent interactions in positronium, muonium, hydrogen, and antiprotonic helium [20].

Antiprotonic helium provides a direct example of a matter–antimatter spin-dependent constraint, having been used to search for exotic spin- and velocity-dependent interactions between electrons and antiprotons [9].

These systems are therefore useful benchmark channels for developing the signed-likelihood methodology advocated here. Where physically possible, future measurements should preserve enough information to construct signed shifts, likelihood functions, or covariance matrices rather than reducing the final result to a positive exclusion magnitude.

XII. DISCUSSION

The main result of this work is methodological. A numerical difference between matter and antimatter exclusion limits is not, by itself, a measurement of CPT violation.

The result follows directly from Eq. (30). If two experiments report positive upper limits, their difference can arise entirely from relative experimental sensitivity. This remains true even if the underlying interaction is exactly CPT conserving.

The compiled registry gives this mathematical observation an empirical context. Only 22 of 247 analysed records involve an antimatter sector, and seven of twelve classified potential groups have no antimatter coverage. The largest sector deficits occur in electron–nucleus, neutron–nucleus, and proton–nucleus interactions.

The numerical values of $|A_U|$ should consequently be described as sensitivity-gap diagnostics. Values in the range

$$|A_U| \simeq 0.8\text{--}1 \quad (54)$$

do not constitute anomalous CPT signals merely because they are numerically large.

The reinterpretation does not diminish the value of antimatter experiments. Instead, it specifies what additional information is required before their results can become quantitative CPT tests. The experimentally useful

objective is to produce matched matter and antimatter measurements of the same operator over overlapping interaction ranges.

The recent literature demonstrates continued progress on both the matter and antimatter sides. Spin sensors and NV-diamond experiments have improved laboratory constraints [11–13], while antiprotonic helium and positronium provide antimatter systems with well-developed precision spectroscopy [9, 10]. New torsion-balance proposals are projected to extend the sensitivity to additional spin-dependent structures [24]. The principal missing ingredient for a broad CPT programme is therefore not another numerical comparison of unequal upper limits, but matched measurements with compatible definitions.

There is also a broader methodological lesson for global compilations of new-physics constraints. A collection of exclusion curves is not automatically a collection of statistically independent measurements. Duplicate records, combined potential classes, correlated interpolation points, different coupling normalisations, and heterogeneous statistical conventions must be retained explicitly.

XIII. LIMITATIONS

The numerical registry is a curated compilation rather than a guarantee of exhaustive coverage of every relevant publication available by the date of this manuscript. The literature continues to develop rapidly, and future releases should incorporate additional measurements and proposals.

Thirty-six of the 283 compiled records are held out of the physics analysis. This is preferable to assigning them unsupported potential labels, but it means that the registry is not completely closed under all possible source conventions.

Several range entries are affected by probable unit-token errors. These entries must be corrected against their source data before the registry is used for global range statistics.

The SME–nonrelativistic matching is complete for the spin-dependent sector of the minimal SME, but only through third order in $1/m$ and for a single fermion species. Higher orders, the spin-independent coefficients a_μ , $c_{\mu\nu}$ and e_μ , and the nonminimal SME are not treated.

The DM labels in the database are phenomenological labels and are not treated as one-to-one images of SME coefficients. A complete relativistic–nonrelativistic–two-body dictionary would require explicit matching of relativistic scattering amplitudes for the underlying mediator models.

Finally, the bound-asymmetry analysis is not a CPT hypothesis test. The autocorrelation analysis can diagnose the effect of correlated interpolation, but it cannot convert one-sided exclusion curves into signed measurements or reconstruct experimental likelihoods that were

not published.

XIV. CONCLUSIONS

We have developed a unified framework for connecting the spin-dependent relativistic SME fermion operators to their controlled nonrelativistic structures and for assessing what can and cannot be learned from matter–antimatter comparisons of published spin-dependent interaction limits.

The canonical FW treatment gives, through the leading orders relevant here,

$$H_{\text{NR}}^{(b)} = -b_i \sigma^i + \frac{b_0}{m} \boldsymbol{\sigma} \cdot \mathbf{p}, \quad (55)$$

$$H_{\text{NR}}^{(H)} = \mathbf{B}_H \cdot \boldsymbol{\sigma} - \frac{1}{m} \boldsymbol{\sigma} \cdot (\mathbf{p} \times \mathbf{H}_E), \quad (56)$$

$$H_{\text{NR}}^{(d)} = m d_{i0} \sigma^i - (d_{ij} + d_{00} \delta_{ij}) p_j \sigma^i, \quad (57)$$

$$H_{\text{NR}}^{(g)} = -m \tilde{\mathbf{g}} \cdot \boldsymbol{\sigma} - \boldsymbol{\sigma} \cdot (\mathbf{p} \times \mathbf{g}_{00}) - \frac{1}{2} \epsilon_{klm} g_{mlj} p_j \sigma^k. \quad (58)$$

These four coefficients are the complete set that generates spin-dependent terms in the minimal SME at these orders.

The d_{i0} and $g_{[kl]0}$ contributions are mass enhanced and agree with the established nonrelativistic SME Hamiltonian. The momentum-linear d sector is governed by the canonically normalised combination $d_{ij} + d_{00} \delta_{ij}$; the earlier treatment in terms of separate unresolved signs is therefore superseded.

The particle–antiparticle relation must likewise be obtained from the complete Hamiltonian. In the canonical SME treatment the relevant coefficient substitutions include

$$\begin{aligned} b_\mu &\rightarrow +b_\mu, & d_{\mu\nu} &\rightarrow -d_{\mu\nu}, \\ g_{\lambda\mu\nu} &\rightarrow +g_{\lambda\mu\nu}, & H_{\mu\nu} &\rightarrow -H_{\mu\nu}, \end{aligned} \quad (59)$$

for the antiparticle Hamiltonian [8]. The CPT classification of a relativistic coefficient should not be confused with the sign of an individual experimentally defined non-relativistic observable.

The principal statistical result is exact. If two positive upper limits are represented by

$$U_f = s_f g_\star, \quad U_{\bar{f}} = s_{\bar{f}} g_\star, \quad (60)$$

then

$$A_U = \frac{s_f - s_{\bar{f}}}{s_f + s_{\bar{f}}}. \quad (61)$$

The underlying coupling cancels, so the statistic is a measure of relative experimental sensitivity rather than a CPT observable. Large values of $|A_U|$ can occur under exact CPT symmetry, while small values do not by themselves establish CPT invariance.

The compiled registry contains 283 records, 247 of which enter the present analysis. Twenty-two involve antimatter, and fifteen pairs (thirteen independent) satisfy the adopted criteria for direct matter–antimatter comparison. Seven of the twelve classified potential groups have no antimatter record. The largest sector deficits occur in electron–nucleus, neutron–nucleus, and proton–nucleus interactions.

The appropriate next-generation programme is therefore two-stage. First, existing exclusion curves should be used to identify balanced sensitivity regions through

$$R_\alpha(\lambda) = \frac{U_{\bar{f},\alpha}(\lambda)}{U_{f,\alpha}(\lambda)}. \quad (62)$$

Second, experiments in those regions should report signed observables or likelihood functions so that the actual CPT hypothesis

$$\theta_f - \eta_{\text{CPT}}\theta_{\bar{f}} = 0 \quad (63)$$

can be tested directly.

XV. FURTHER WORK

Several extensions follow naturally.

First, the canonical reduction should be carried to higher order in $1/m$, where the spin-dependent sector acquires further structure, and extended to the spin-independent coefficients a_μ , $c_{\mu\nu}$ and e_μ , which bear on the V_1 channel.

Second, the one-body SME structures should be matched explicitly to two-fermion relativistic amplitudes for specified mediator models. This is the appropriate route to a genuine SME–DM coupling dictionary.

Third, the constraint registry should preserve source likelihoods, confidence conventions, covariance information, coupling normalisations, and experimental provenance whenever these are available. Such a structure

would allow future matter and antimatter measurements to be compared without reconstructing the statistical problem from published exclusion curves.

Finally, the most informative future experiments are those that produce matched matter–antimatter observables for the same operator over overlapping interaction ranges. Once such measurements exist, the sensitivity-ratio analysis developed here can identify the relevant coverage, while a likelihood-level comparison can provide the actual quantitative CPT test.

DATA AVAILABILITY

The constraint registry, the matched-pair results, the analysis pipeline (SPINDEP) and the figures generated from them are openly available at <https://github.com/oyewodayo/spindep-framework>. The symbolic Foldy–Wouthuysen derivations, including the check of the particle–antiparticle relation for every minimal-SME coefficient family, are available at <https://github.com/oyewodayo/spin-dependent-exotic-interactions-SME>. The numerical results in Tables II–V are regenerated from the registry by a single pipeline command.

ACKNOWLEDGMENTS

We thank the experimental and theoretical communities whose published results and constraint data provide the foundation for systematic cross-platform comparisons of exotic spin-dependent interactions. TSO gratefully acknowledges CERN for sponsoring his participation in the CERN Summer Student Lecture Programme 2026 and for the scientific training and research environment provided during his participation. He also acknowledges the CMS Collaboration for the broader scientific training and collaborative research environment which supported the development of this work.

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